

FAAR to AAI

Verified Findings and Execution Status

Evidence report generated from the FAAR repository

8 August 2026

Scope warning. This is a FAAR report for the repository under review. It is not the unrelated ACOM paper. All numbers below are copied from repository artifacts or the run log. A number is not treated as a paper result unless the underlying run is a real planned experiment with a saved result artifact.

1 Executive summary

The repository is on branch `faar-aaai-experiments`. It contains two related but different evidence layers:

1. The prototype layer contains a reproducible, offline, 40-example mock evaluation and the implemented FAAR controller.
2. The AAI execution layer contains the locked protocol, model and dataset provenance, asset preparation, smoke testing, and commands for the planned B0 to B4 baselines.

The central result of this audit is:

The planned AAI baselines B0, B1, B2, B3, and B4 have produced **zero of five paper-eligible result files**. The only B0 file is explicitly a one-document smoke test and is marked `paper_result=false`.

The prototype slice does contain numbers. On the current 40-example offline mock profile run, every compared profile has $EM = 0.1250$ and $F1 = 0.2036$. FAAR has not shown a quality advantage over `naive.rag` on this slice: the recorded deltas are zero. The same artifacts support selective routing and diagnosis-policy consistency, but they do not support a broader answer-quality improvement claim.

The remaining work is therefore experimental execution, not another paper rewrite: complete the GPU-backed assets, finish ArXivQA page confirmation and OCR, obtain MP-DocVQA, run B0 to B4 in order with B2 included, and only then populate paper tables.

2 How the repository was interpreted

The apparent contradiction between earlier progress wording and the actual baseline state comes from two phase vocabularies in the same FAAR repository.

Table 1: Evidence layers that must not be conflated.

Layer	What it contains	How it is used in this report
Prototype	The earlier FAAR controller, formal method mapping, 40-example benchmark profiles, phase reports, and claim artifacts.	Reported as offline prototype evidence. It is useful for engineering and hypothesis checking, but it is not substituted for the planned AAI B0 to B4 runs.

Layer	What it contains	How it is used in this report
AAAI execution	The fixed OHR split, pinned model revisions, asset contracts, full-paper ArXivQA preparation, B0 smoke path, and the ordered B0 to B4 protocol.	Reported as readiness and execution status. A planned baseline counts only when its real result file and provenance exist.

The older prototype report says that Phases 0 to 4 are complete, while its benchmark command uses a mock VLM and `--no-api-enabled`. The AAAI execution plan separately states that no Phase 1 baseline has started. This report preserves both facts and labels them by layer instead of merging them.

Primary evidence: `PLAN.md`, `RUNLOG.md`, and the supervisor report listed in the evidence index.

3 Phase and plan status

3.1 AAAI execution track

Table 2: AAAI execution status at the report date.

Phase	Status	Verified work	Paper run
0 Setup	Implementation complete; asset readiness incomplete	Fixed OHR split, dataset registration, pinned loaders, shared-corpus guards, Docling and GOT-OCR preparation, API accounting, evaluation fields, model provenance, and one-document smoke path.	Smoke only
1 Baselines	Protocol implemented; execution pending	Ordered B0, B1, B2, B3, B4 commands are recorded. No full baseline result file exists.	0 of 5
2 Quality gate	Implementation complete; real validation pending	BGE top-1 threshold search and lock-file machinery exist. A real B0 validation result is required before choosing a threshold.	Not started
3 Annotation	Tooling complete; human study pending	Failure sampling, GOT-OCR extraction, Label Studio assets, and kappa verification are implemented.	Not started
4 Main FAAR runs	Implementation complete; assets and credentials pending	Dataset and split provenance, complete-asset guards, runner, and cost logging exist.	Not started
5 Ablations	Implementation complete; experiment inputs pending	A1 to A4 routing variants and tests exist.	Not started
6 Analysis	Implementation complete; real result JSONs pending	Bootstrap intervals, failure-type analysis, harm rate, and Pareto export are implemented.	Not started
7 Tables and submission checks	Generation tooling complete; inputs pending	Table and figure generation plus anonymization checks exist. No final paper PDF is present.	Not started

The phase statuses above come from `PLAN.md:25–37`, `PLAN.md:39–71`, `PLAN.md:73–119`, and the latest entries in `RUNLOG.md:61–69`. “Complete” in the implementation column means code and contracts are present; it does not mean that the corresponding paper experiment has been executed.

3.2 Prototype track

The earlier prototype phases are operationally useful and have committed evidence:

- Prototype Phase 0: a fixed 40-example grounding set with 33 no-issue, 5 text-corruption, and 2 structure-corruption examples.
- Prototype Phase 1: a working LangGraph controller with typed recovery and a completed one-example end-to-end prototype log.
- Prototype Phase 2: formal definitions and code-to-math mapping for Q , F , π , and U .
- Prototype Phase 3: profile runner and metric exports for `faar_full`, `faar_no_backtrack`, `faar_no_vlm`, `faar_no_diagnosis`, and `naive_rag`.
- Prototype Phase 4: claim assessment and case-study artifacts.
- Prototype Phase 5: claim-safe writing scaffolding.

These are prototype implementation milestones. They are not evidence that the AAAI B0 to B4 protocol has been run.

4 Verified data and asset findings

4.1 OHR-Bench split and smoke test

Table 3: Locked OHR-Bench counts and observed preparation costs.

Item	Questions	Documents	Pages	Evidence
Train split	5,948	—	—	RUNLOG.md:8
Validation split	1,274	549	7,033	RUNLOG.md:50
Test split	1,276	567	6,842	RUNLOG.md:50
One-document smoke	1	1	1	RUNLOG.md:57, RUNLOG.md:64

The selected smoke document is `academic/DUDE.157911e3080d18f4d799a122aaeb33fb`. The observed one-page run used MPS and recorded Docling time of 8.91 seconds, rendering time of 0.85 seconds, GOT-OCR time of 1,484.05 seconds per page, peak RSS of 1.72 GiB, a 609,929-byte PNG, a 7,100-byte OCR file, and an approximately 2.12 GiB GOT-OCR cache. A linear extrapolation recorded in the run log gives approximately 2,899 OCR-hours for the validation pages and 2,821 OCR-hours for the test pages. These are planning estimates, not completed runtimes.

The B0 smoke used one example, 14 chunks from real GOT-OCR text, lexical mock retrieval, no paid API request, and no paper-result aggregation. An earlier intermediate asset-load note records 48 chunks for the same one-document preparation path. The difference is an intermediate corpus accounting detail and is not used as an experimental result.

4.2 ArXivQA full-paper remapping preparation

Table 4: ArXivQA preparation evidence.

Quantity	Verified value
Staged QA rows	500
Distinct papers	497
Rendered paper pages	8,955
Candidate mappings accepted	0
Candidate mappings unresolved	500
OCR entries in paper inventory	0
Inventory state	<code>preparation_only</code>
Paper-ready flag	<code>false</code>

The staged subset was matched against the official 100,000-row JSONL using image, question, answer, and normalized options. The exact source locks are recorded below.

Official revision	85a6dca0e2bdc6f0268ae519be8913f83a83cafd
Official SHA-256	d66cee08f152747ea58ce7851310c97229dcc6103901b3041ea988ad11eb1b96
ViDoRe revision	b8a106812c8682bab08935cf5d1b4566c82562de
ViDoRe SHA-256	a45cb85bb49dce83b1beb851e312fe38179035151a55401a8801a2e53c1159d3
Staged-row fingerprint	ccb60459021d211628c399682437841158e1ca238799a3cd81eef63026a7fa9a

The withdrawn paper 2304.04203 has an explicit v1 override. Its recorded v1 match score is 0.7699745893478394, while the maximum v2 score is 0.2556701898574829. This is provenance for an asset override, not a model evaluation score.

The exact source and readiness fields are in the following files:

- `config/arxivqa_source_lock.json`
- `data/benchmark_prep/arxivqa/staging_manifest.json`
- `data/benchmark_prep/arxivqa/paper_inventory.json`
- `data/benchmark_prep/arxivqa/evidence_candidates.json`

The latest run-log summary is `RUNLOG.md:66-69`.

4.3 Other data blockers

MP-DocVQA validation is not present locally. The run log records the expected validation scale as 5,187 questions, 927 documents, and 6,510 pages, with download access requiring the authenticated RRC DocVQA Task 4 portal. Complete OHR OCR and final ArXivQA OCR also require GPU compute. The OpenAI baseline calls have not started.

5 Baseline numbers

5.1 Planned AAI baselines B0 to B4

Table 5: Paper-eligibility audit for the planned baselines.

ID	Expected output	Verified status
B0	<code>results/b0.json</code>	Absent. Only <code>results/smoke/b0_one_doc.json</code> exists, and it is explicitly a one-document smoke test.
B1	<code>results/b1.json</code>	Absent. No always-VLM baseline result exists.
B2	<code>results/b2.json</code>	Absent. No random-recovery baseline result exists. This required ablation is not skipped; it is still pending.
B3	<code>results/b3.json</code>	Absent. No ColPali baseline result exists.
B4	<code>results/b4.json</code>	Absent. No VisRAG baseline result exists.
Total	Five planned paper baselines	0 of 5 completed

The ordered commands are recorded in `docs/faar-aaai-plan.md:47-59`, and the absence of the five files was checked in the repository result directory. Earlier blocked attempts are recorded in `RUNLOG.md:12-16`. They did not create result files and are not treated as results.

5.2 B0 one-document smoke numbers

The values above are from `results/smoke/b0_one_doc.json`. The file also records the fixed model names and revisions, but the smoke path did not perform a paid VLM call or a full pinned-model baseline run. It must not be placed in a paper table beside B1 to B4.

Table 6: B0 smoke payload, reported only as a smoke test.

Field	Value
Evaluation examples	1
EM	0.0
F1	0.0
VLM rate	0.0
Harm rate	0.0
API requests	0
Prompt tokens	0
Completion tokens	0
Cost in USD	0.0
Runtime in seconds	0.0226
Mock retrieval	true
API enabled	false
Paper result	false

5.3 Prototype offline mock profile numbers

The following table is real repository output, but it is a separate offline mock profile evaluation with $n = 40$ per profile. It is not B0 to B4.

Table 7: Offline 40-example mock profile evaluation.

Profile	N	NDCG@5	Recall@5	EM	F1	VLM fallback
<code>faar_full</code>	40	0.4789	0.5250	0.1250	0.2036	0.0750
<code>faar_no_backtrack</code>	40	0.4789	0.5250	0.1250	0.2036	0.0750
<code>faar_no_diagnosis</code>	40	0.4789	0.5250	0.1250	0.2036	0.0000
<code>faar_no_vlm</code>	40	0.4789	0.5250	0.1250	0.2036	0.0000
<code>naive_rag</code>	40	0.4789	0.5250	0.1250	0.2036	0.0000

Source: `artifacts/phase3/metrics_summary.json` and `artifacts/phase4/profile_deltas_vs_baseline.csv`. The supervisor report records a rerun with 40 examples, seed 42, a mock VLM backend, and API calls disabled. The profile deltas versus `naive_rag` are zero for EM, F1, NDCG@5, and Recall@5 for every profile.

5.4 Prototype routing and claim findings

For `faar_full` on the same 40-example mock slice, the recorded routing distribution is:

Route or failure type	Count	Rate
Pass / answer direct	21	0.525
Word level / correct text	14	0.350
Structural / invoke VLM	3	0.075
Semantic / retry retrieval	2	0.050
Non-pass total	19	0.475

The claim artifact marks the following as supported or unsupported:

- Selective fallback cost behavior: supported in this mock profile artifact; visual fallback rate = 0.075, with an estimated 0.925 savings relative to an always-on visual profile.
- Diagnosis-policy consistency: supported; recorded consistency = 1.0.

- Typed-recovery quality gain over `naive_rag`: not supported; delta F1 = 0.0 and delta EM = 0.0.
- Selective visual fallback quality gain over no VLM: not supported; delta F1 = 0.0 and delta EM = 0.0.

These are conservative prototype findings. The supervisor report explicitly states that the command used a mock visual backend with API calls disabled, so the results do not establish performance of the full multimodal AAAI stack.

6 Implementation changes and provenance

The current AAAI change set added or hardened the following repository capabilities:

- locked OHR split and dataset registration;
- complete-document retrieval contracts that prevent evidence pages from defining the retrieval corpus;
- exact model revision recording for NV-Embed-v2, BGE reranker, GOT-OCR, ColPali, VisRAG, and ByT5;
- Docling and pinned GOT-OCR asset preparation with resumable provenance;
- OpenAI request, token, cost, request-model, response-model, and completion-time logging;
- strict full-paper ArXivQA remapping with source fingerprints, version overrides, PDF and image hashes, and human-confirmed mapping requirements;
- one-document B0 smoke execution and explicit smoke-versus-paper-result fields;
- phase 2 gate tuning, phase 3 annotation, phase 4 main-run, phase 5 ablation, phase 6 analysis, and phase 7 table-generation tooling;
- tests for routing, provenance, path safety, shared-corpus integrity, ArXivQA preparation, remapping, and CLI behavior.

The most recent ArXivQA-related commits are:

Table 8: Recent committed changes on the FAAR experiment branch.

Commit	Message	Purpose
17128f0	Ignore external benchmark inputs	Keep downloaded inputs outside versioned source.
5ffffe18	Lock ArXivQA remapping sources	Record exact source revisions and hashes.
78ad6f2	Prepare full-paper ArXivQA assets	Add preparation workflow and inventory support.
fbab33e	Add ArXivQA preparation CLI	Expose repeatable staging and preparation commands.
2bd47e8	Harden ArXivQA remapping provenance	Fail closed on source and mapping inconsistencies.
d4bc236	Cover ArXivQA asset preparation	Add preparation and provenance tests.
2791339	Enforce full-paper remap integrity	Protect complete-paper retrieval and mapping contracts.
b69fb2b	Record ArXivQA preparation evidence	Document asset counts, status, and blockers.
94e27a6	Refresh phase zero asset status	Update readiness and execution status.
8a7ef92	Ignore local build artifacts	Keep generated build output out of source control.

The branch was clean and matched `origin/faar-aaai-experiments` before this report was added. The full history remains available through Git; the table highlights the latest experiment-specific changes rather than claiming that these are the only changes in the repository.

7 Verification status

The latest repository run log records the following engineering checks:

Table 9: Verification evidence and limits.

Check	Recorded result	Interpretation
Locked Python test suite	136 passed, 4 warnings	Current committed implementation regression check passed.
Locked dependency check	<code>pip check</code> passed	The locked environment has consistent declared dependencies.
Compilation and package build	Passed in run log	Source compiles and a wheel was previously built.
Split integrity	<code>RUNLOG.md:28, 47, 59</code>	The immutable split was not changed.
Credential tracking check	No credential files tracked	No API secret was added to Git.
Coverage	No committed coverage result	Coverage is not a verified project result yet.
Dead-code check	No configured dead-code report	No deletion should be justified from an unconfigured tool.

The verification evidence is recorded in `RUNLOG.md:65-69`, `results/environment/pip-freeze.txt`, and `pyproject.toml`. The repository has a GitHub Actions pytest workflow, but a passing local test suite is not the same as a completed paper evaluation.

7.1 Requested external worker audit

Five read-only OpenCode workers were launched with the requested main, non-free DeepSeek Flash model. The first attempt was blocked by the local sandbox network boundary. A retry with network permission reached the requested service, but all five jobs were rejected with an “Insufficient balance” error before producing review output. The free model was not used, and no worker output was incorporated as evidence. The repository audit in this document is therefore based on the checked-in artifacts and direct local commands.

8 What remains

The next execution sequence is:

1. Choose or provision a GPU workflow for complete OHR val/test page rendering and GOT-OCR. The observed MPS smoke rate is not a practical full-dataset plan.
2. Finish the 500 ArXivQA QA-to-paper/page confirmations. Keep every mapping human-confirmed and retain complete-paper retrieval corpora.
3. Run pinned GOT-OCR for the 8,955 ArXivQA pages and emit per-page provenance before registering the dataset.
4. Download the authenticated MP-DocVQA validation set from the RRC Task 4 portal and record checksums.
5. Run the planned baselines in order: B0, B1, B2, B3, and B4. Save each result before starting the next. B2 must be included.
6. Tune the gate on validation data only after a real B0 result exists, then lock the threshold.
7. Run the human annotation study, FAAR test runs, ablations, confidence intervals, failure-type analysis, and paper tables.

Until Step 5 is complete, the correct supervisor statement is: “The FAAR implementation and experiment controls are substantially prepared, but the paper-eligible B0 to B4 baseline results are still pending because full OCR assets, ArXivQA confirmation/OCR, MP-DocVQA access, and paid baseline execution are not complete.”

9 Evidence index

Table 10: Primary files used for this report.

File	Use
PLAN.md	AAAI phase status, implementation-versus-execution distinction, and remaining blockers.
docs/faar-aaai-plan.md	Supervisor-approved ordered B0 to B4 commands and checklist.
RUNLOG.md	Timestamped execution evidence, smoke numbers, asset counts, test counts, and blockers.
results/smoke/b0_one_doc.json	The only B0-labelled result file; explicitly non-paper smoke payload.
artifacts/phase3/metrics_summary.json	Current 40-example offline mock profile metrics.
artifacts/phase4/claim_assessment.json	Supported and unsupported prototype claims.
artifacts/phase4/profile_deltas_vs_baseline.csv	Profile deltas against naive_rag .
config/model_revisions.json	Locked Hugging Face model repositories and revisions.
config/arxivqa_source_lock.json	ArXivQA source revisions, hashes, and withdrawn-paper override.
data/benchmark_prep/arxivqa/paper_inventory.json	ArXivQA rendered-page inventory and OCR readiness state.
data/benchmark_prep/arxivqa/evidence_candidates.json	Mapping candidate status; all 500 remain unresolved.
docs/reports/supervisor_progress_report.md	Prototype mock command and the earlier supervisor-facing interpretation.
docs/phases/phase5/writing_plan.md	Claim-safe writing guidance and limits on current evidence.